

Test Cases

#	Input	Expected Output	Actual Output	Pass/ Fail
1	connecting to 192.168.1.1 now	Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)	Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)	PASS
2	server=10.0.0. 255:8080end	Extracted IPv4 address: 10.0.0.255 (decimal value: 167772415, port: 8080)	Extracted IPv4 address: 10.0.0.255 (decimal value: 167772415, port: 8080)	PASS
3	192a168.1.1.1	Extracted IPv4 address: 168.1.1.1 (decimal value: 2818638081, port: none)	Extracted IPv4 address: 168.1.1.1 (decimal value: 2818638081, port: none)	PASS
4	192.168.1.1.	Invalid input: no valid IPv4 address found	Invalid input: no valid IPv4 address found	PASS
5	Connection from 192.168.1.1 refused	Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)	Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)	PASS
6	192.168.01.1	Invalid input: no valid IPv4 address found	Invalid input: no valid IPv4 address found.	PASS
7	1.2.3.4:99999	Invalid input: no valid IPv4 address found	Invalid input: no valid IPv4 address found.	PASS
8	12.34.56	Invalid input: no valid IPv4 address found	Invalid input: no valid IPv4 address found.	PASS
9	no number here	Invalid input: no valid IPv4 address found	Invalid input: no valid IPv4 address found.	PASS
10	END	Program terminated. (loop exits)	Program terminated.	PASS
11	END 1.1.1.1	Extracted IPv4 address: 1.1.1.1 (decimal value: 16843009, port: none)	Extracted IPv4 address: 1.1.1.1 (decimal value: 16843009, port: none)	PASS
12	0.0.0.0	Extracted IPv4 address: 0.0.0.0 (decimal value: 0, port: none)	Extracted IPv4 address: 0.0.0.0 (decimal value: 0, port: none)	PASS
13	255.255.255.2 55:65535	Extracted IPv4 address: 255.255.255.255 (decimal value: 4294967295, port: 65535)	Extracted IPv4 address: 255.255.255.255 (decimal value: 4294967295, port: 65535)	PASS
14	192.168.1.1.5	Invalid input: no valid IPv4 address found.	Invalid input: no valid IPv4 address found.	PASS
15	192..1.1	Invalid input: no valid IPv4 address found.	Invalid input: no valid IPv4 address found.	PASS
16	1.2.3.4:80:90	Invalid input: no valid IPv4 address found.	Invalid input: no valid IPv4 address found.	PASS

#	Description
1	Basic extraction of a valid address within surrounding text (No port)
2	Valid address with a valid port within text with a trailing “end” at the end
3	The letter “a” separating “192” from the valid address “168.1.1.1”
4	Trailing “.” makes the address “192.168.1.1.” invalid, resulting in a no address found message
5	Same as #1
6	Leading-zero “01” within the third octet, resulting in a no address found message
7	“99999” exceeds the port range of 65535, resulting in a no address found message
8	Only 3 octets found, resulting in a no address found message
9	No digits at all, resulting in a no address found message
10	The exact string “END” should end the main loop and terminate the program.
11	Program should still extract the address “1.1.1.1” even with the “END” at the very beginning
12	Same as #1
13	Test the max ranges of the octet and ports
14	Extra 5th octet in the address, resulting in a no address found message, resulting in a no address found message
15	Empty octet within the address
16	Extra port octet in the address, resulting in a no address found message, resulting in a no address found message